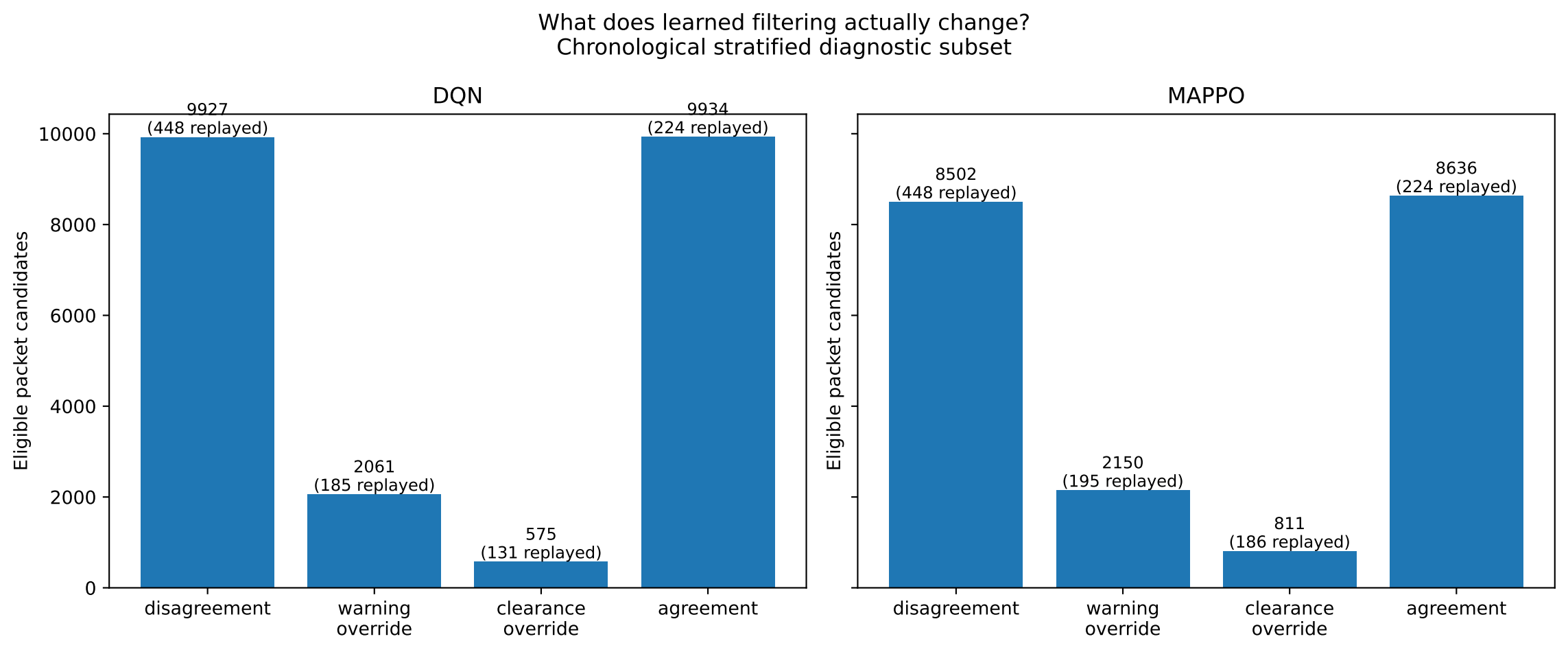
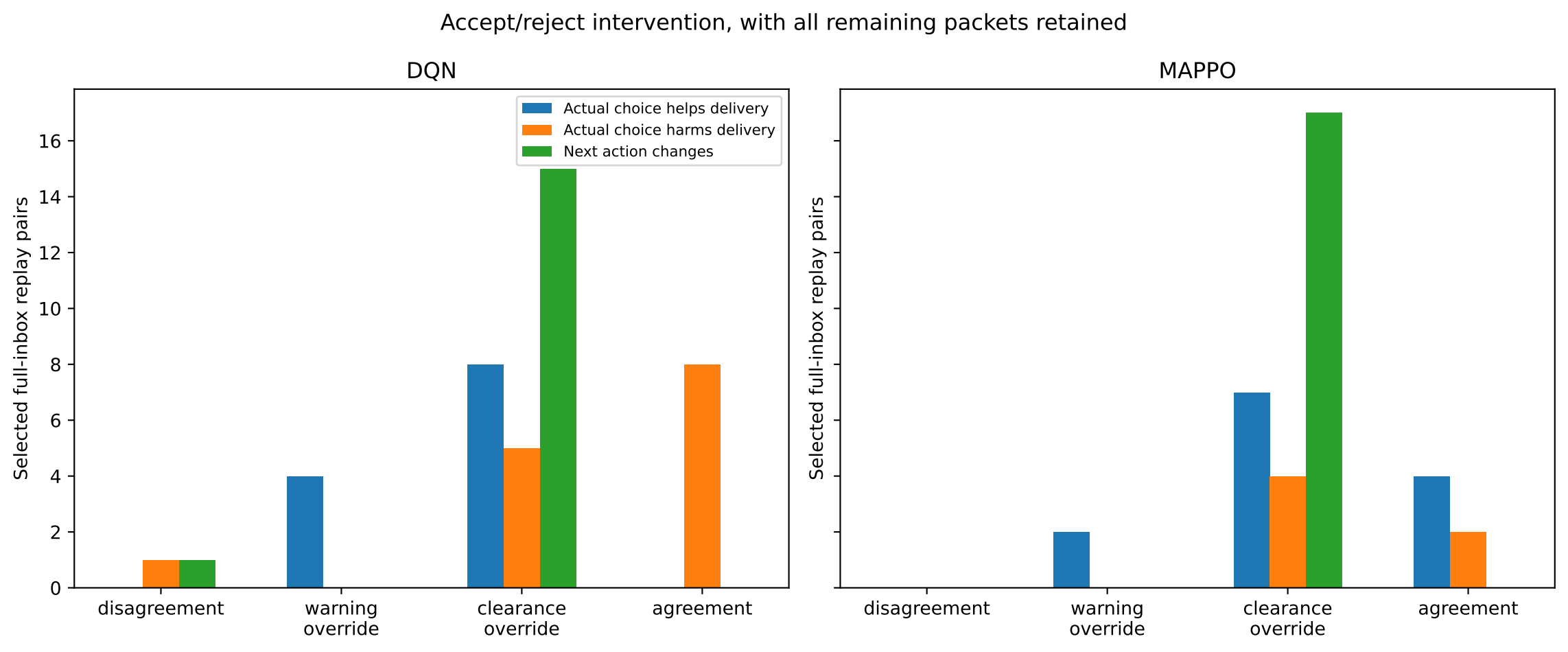
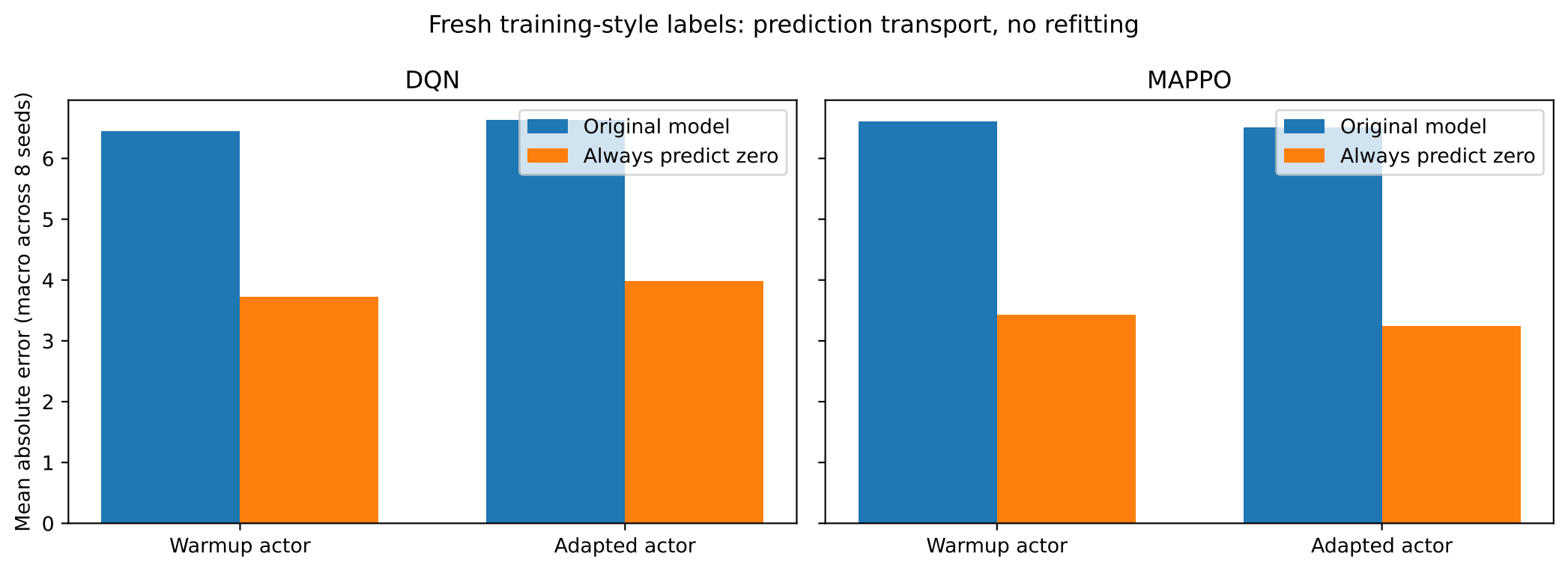
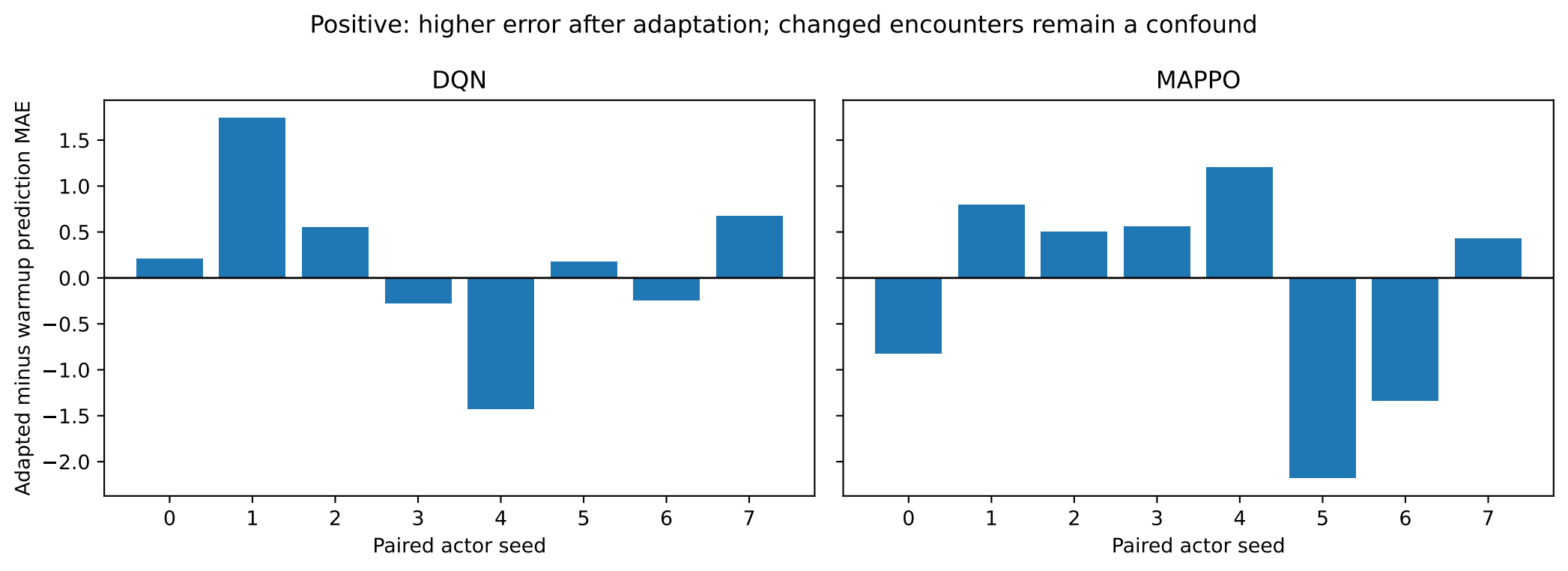
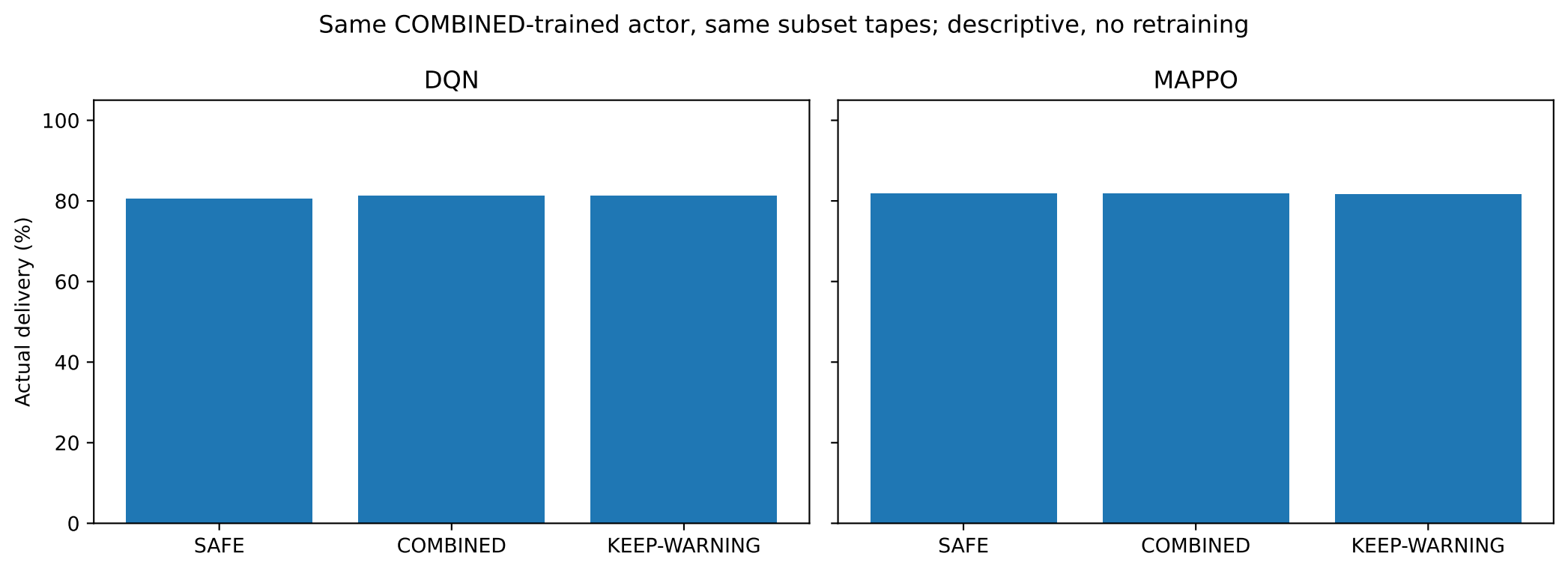
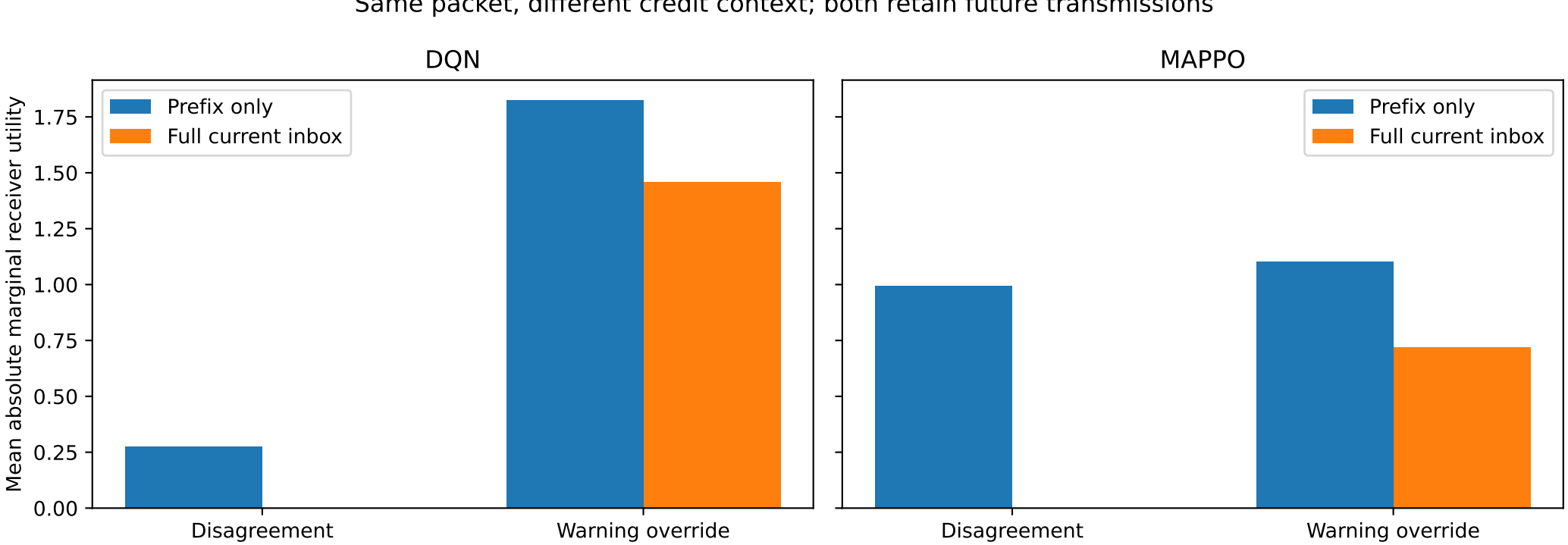

What does learned filtering actually change?
Chronological stratified diagnostic subset
DQN
MAPPO
Eligible packet candidates
0
2000
4000
6000
8000
10000
9927
(448 replayed)
2061
(185 replayed)
575
(131 replayed)
9934
(224 replayed)
8502
(448 replayed)
2150
(195 replayed)
811
(186 replayed)
8636
(224 replayed)
disagreement
warning override
clearance override
agreement

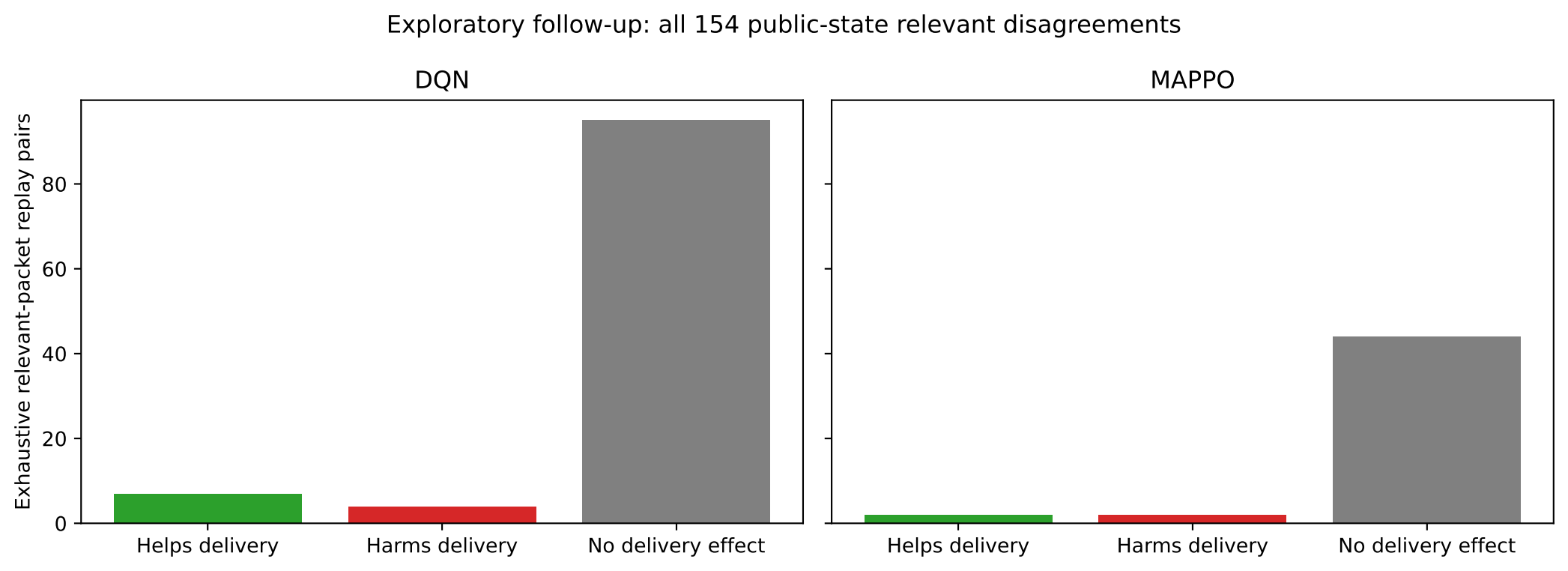

Accept/reject intervention, with all remaining packets retained
DQN
MAPPO
Actual choice helps delivery
Actual choice harms delivery
Next action changes
Selected full-inbox replay pairs
0
2
4
6
8
10
12
14
16
disagreement
warning override
clearance override
agreement


Fresh training-style labels: prediction transport, no refitting
DQN
MAPPO
Mean absolute error (macro across 8 seeds)
0
1
2
3
4
5
6
Original model
Always predict zero
Warmup actor
Adapted actor


Positive: higher error after adaptation; changed encounters remain a confound
DQN
MAPPO
Adapted minus warmup prediction MAE
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
1
2
3
4
5
6
7
Paired actor seed
Paired actor seed


Same COMBINED-trained actor, same subset tapes; descriptive, no retraining
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
SAFE
COMBINED
KEEP-WARNING
SAFE
COMBINED
KEEP-WARNING


Same packet, different credit context; both retain future transmissions
DQN
MAPPO
Prefix only
Full current inbox
Mean absolute marginal receiver utility
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Disagreement
Warning override


Exploratory follow-up: all 154 public-state relevant disagreements
DQN
MAPPO
Exhaustive relevant-packet replay pairs
0
20
40
60
80
Helps delivery
Harms delivery
No delivery effect
Helps delivery
Harms delivery
No delivery effect